

BE 507: Tissue Engineering

Course Code	BE 507
Course Name	Tissue Engineering
L-T-P-C	3-0-0-3
Prerequisites	IC 136 Understanding Biotechnology and its Applications or IC 241 Materials Science for Engineers or Consent of Faculty Member
Intended for	B.Tech.-M.Tech. Integrated Dual Degree in Bio-Engineering, M. Tech Biotechnology and PhD candidates
Mutual Exclusion	NA
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction to Tissue Engineering:** The history and scope of tissue engineering, Challenge in imitating nature, Cells as building blocks, Clinical translation (4 Hours)
- **Cellular differentiation and Tissue Development:** Molecular organization in cells, Cell-matrix interactions, Tissue development (4 Hours)
- **Functional Tissue Engineering:** Matrix as growth factor reservoir, Mechanobiology of matrix, Biosimilar materials as scaffolds (6 Hours)
- **3D Tissue Culture Techniques:** Animal cell culture, Biomaterials in tissue engineering, Cell interactions with polymers, conventional 3D scaffolds, 3D printing, 3D bioprinting (8 Hours)
- **Transplantation of Engineered Tissues & Disease Models:** Host Immune Response, Immunomodulation, Disease models, applications of disease models (6 Hours)
- **Orthopaedic Tissue Engineering:** Mesenchymal stem cells, Bone ultrastructure and anatomy, Bone tissue engineering, Articular cartilage tissue engineering, Intervertebral disc tissue engineering, Orthopaedic disease models (8 Hours)
- **Ophthalmic Tissue Engineering:** Stem cells in the eye, Corneal replacements, Ophthalmic disease models (6 Hours)

Text Books

1. Robert Lanza, Robert Langer and Joseph Vacanti, **Principles of Tissue Engineering**, Academic press, 2020
2. Bikramjit Basu, **Biomaterials Science and Tissue Engineering: Principles and Methods**, Cambridge University Press, 2017.

References

1. Bikramjit Basu, **Biomaterials for Musculoskeletal regeneration: Concepts**, Springer Nature, 2017.
2. Ravi Birla, **Introduction to tissue engineering: applications and challenges**, John Wiley & Sons, 2014.